

Science A–Z



Learning A–Z

level **W**

Grade

5–6

Lexile

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Unit • Light Energy

Multi-level

U V W

Word Count

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Nonfiction • Physical

KEY ELEMENTS USED IN THIS BOOK

The Big Idea: Many aspects of our lives are affected by or dependent upon light. In addition to learning about light, students may gain an appreciation for the extensive roles that light plays in their lives.

Key comprehension skill: Interpret charts, graphs, and diagrams
Other suitable comprehension skills: Main idea and details; compare and contrast; summarize information; identify facts; elements of a genre; classify information

Key reading strategy: Ask and answer questions
Other suitable reading strategies: Connect to prior knowledge; summarize; visualize; make, revise, and confirm predictions

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LIGHT

Written by Brian Roberts



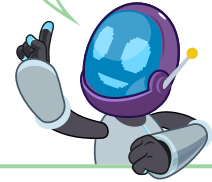
LIGHT



Written by Brian Roberts

Focus Question

What is visible light?



Words to Know

concave lens	photon
convex lens	reflection
fluorescent bulb	refraction
frequency	retina
incandescent bulb	spectrum
laser	translucent
light-year	transparent
medium	wavelength
opaque	white light



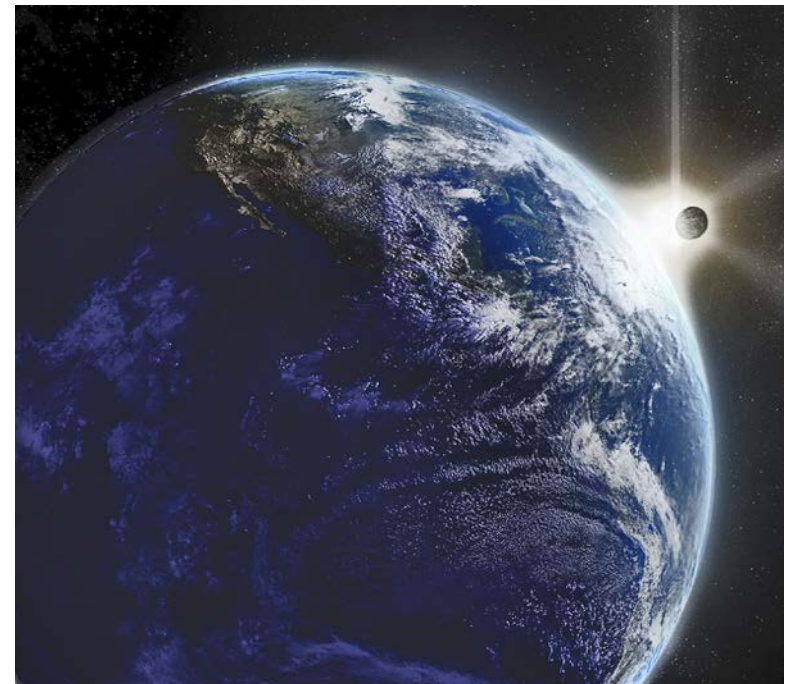
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Introduction

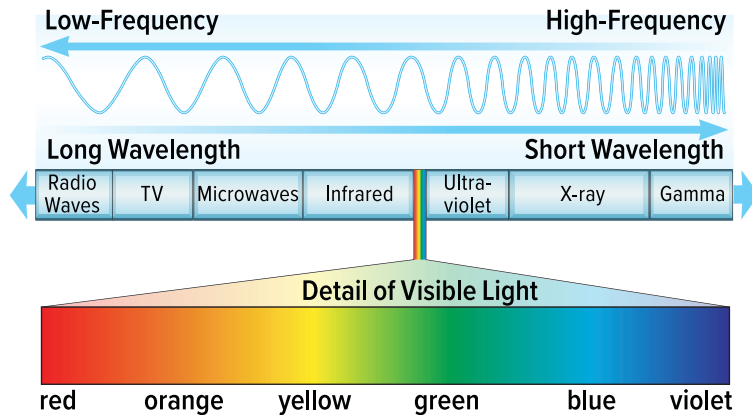
If you've ever walked into a closet and closed the door, you've experienced a world without light. Without light, we can't see a thing. But allowing us to see things is only one aspect of light, especially sunlight.

This book will reveal many of the fascinating facts that scientists have discovered about light. These facts include what it is, where it comes from, how it moves, and what happens when light strikes different surfaces. There is much more to light than meets the eye.



Seen from space, light from the Sun reflects off Earth's oceans.

KINDS OF RADIANT ENERGY



What Is Light?

Light is a form of energy called *radiant energy*. But the light you can see is only one kind of radiant energy. Scientists usually describe the range of radiant energy as a **spectrum**, or a series of energy bands. The diagram above shows all the major bands of radiant energy. Visible light bands are in the middle of this spectrum but are only a small part of it. On each side of visible light are bands representing invisible radiant energy. Those on the left side have less energy than the bands on the right side. For example, gamma rays are much more powerful than radio waves.

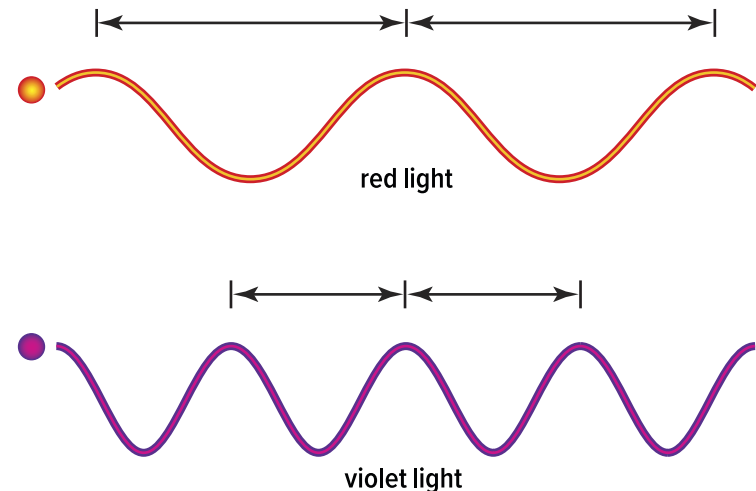
Sunlight is actually a mixture of many colors. Light passing through a prism is separated into many bands of color—the same colors seen in a rainbow.

How Light Moves

Scientists have two ways to describe light and how it moves. The first way describes light as tiny, invisible particles, or packets, of energy called **photons**. Because photons are energy, not matter, they have no weight—even if you gather together billions of them!

Not all photons are equal. High-powered radiant energy, such as gamma rays, has photons that carry more energy than photons of visible light. Even in visible light, one color of light might have more energy than another. For example, violet light photons carry more energy than red light photons.

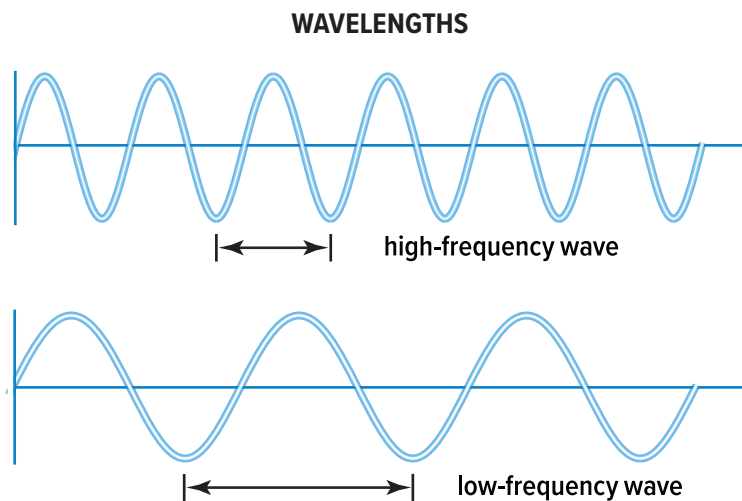
COMPARING COLORS OF LIGHT



Violet light photons have shorter wavelengths and more energy than red light photons.

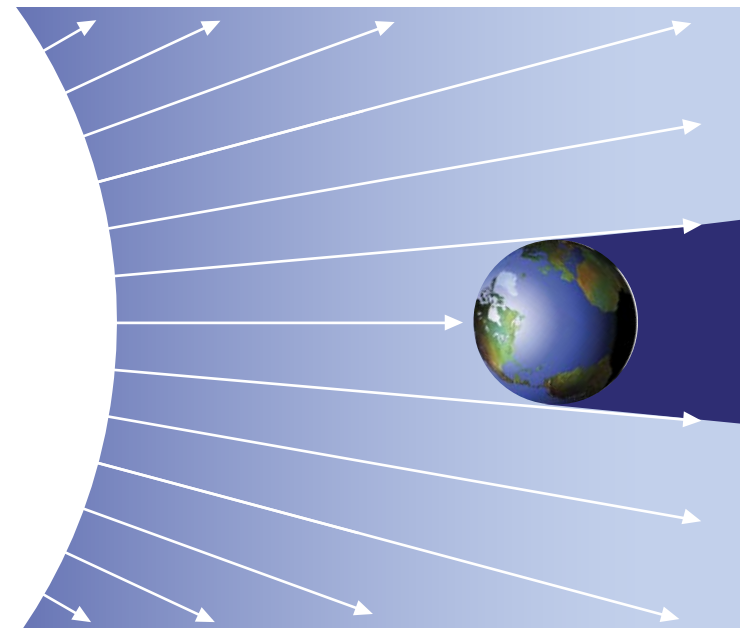
Light is also described as moving in waves. These waves are measured in terms of their **frequency**, or how closely together they travel. High-frequency waves travel more closely together, meaning that more of them pass a given point in a period of time than low-frequency waves, which are more spread out. Scientists measure frequency by a property called **wavelength**. Wavelength is the distance between the peak of one wave and the peak of the wave next to it.

Look at the diagram of a high-frequency wave and a low-frequency wave. The low-frequency waves are more spread out—they have a longer wavelength. The longer the wavelength, the lower the wave's energy.



There are several things you should know about the movement of light. One is that it travels in a straight line. A second is that beams, or rays, of light can change direction. You will read more about this in the section on how light behaves.

A third important thing you should know about light's movement is that, unlike sound, it does not need a **medium** such as air, water, or wood to travel through. This is what makes radiant energy so special—it can travel through the nothingness of space. If it were not for this special property of radiant energy, sunlight would never reach Earth.



Light rays travel in a straight line through space from the Sun to Earth.

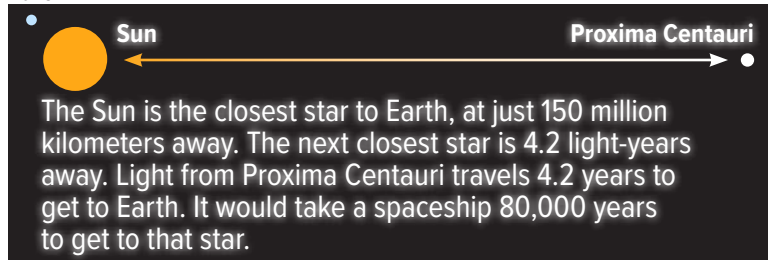
Perhaps the most impressive thing about how light travels is its speed. Nothing is faster. In fact, light moves so fast through space that it travels 300,000 km (186,000 mi.) in a single second. This is much faster than sound travels.



You see lightning before you hear thunder because light travels faster than sound.

Since light travels so fast and over such great distances in space, scientists have come up with a special unit called a **light-year**. It is the distance light travels in one year. That's about 9.5 trillion kilometers (5.9 trillion mi.)!

Earth



**Math
Moment**

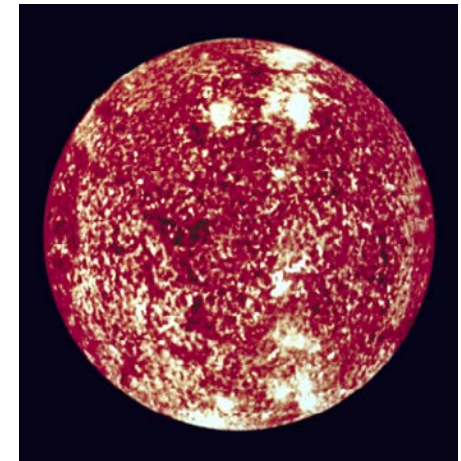
The distance from Earth to the Sun is about 150 million kilometers. Light moves at about 300,000 kilometers per second. About how long does it take light to reach Earth from the Sun?

Where Does Light Come From?

Now that you know what light is and how it moves, let's discuss where it comes from. Sources of light can be placed into two categories—natural and human-made.

The most important natural source of light is the Sun. The radiant energy it sends to Earth is responsible for the food we eat, our weather, most of our electricity and fuel, and the warmth of our atmosphere. But don't be misled into believing the hot Sun sends its heat to Earth. The Sun is too far from Earth for that to happen. Instead, the radiant energy from the Sun is absorbed by Earth's surface. Only then does this radiant energy change to heat energy.

Remember that the Sun is just one of billions of stars in the universe giving off radiant energy. The fact that the Sun is much closer to Earth than any other star makes it possible for us to get so much energy from it.



The Sun is a giant ball of liquid energy and gases. It sends light in every direction.



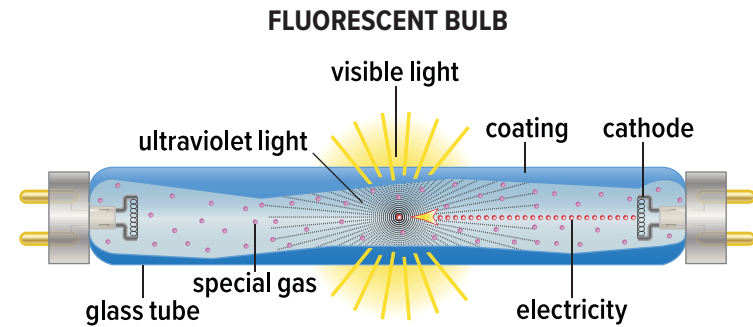
Lava and fireflies are two natural sources of light.

Other sources of natural light exist as well. Take lightning, for example. It is nothing more than super-charged particles of air that get so hot, they produce light energy. Some volcanoes produce super-hot lava that gives off light. Animals like fireflies and glowworms produce light, too.

But humans wanted light where there was little or no natural light. So they invented ways to produce light. At first, it was simple things like campfires and oil lamps that burned with a flame. Then a man named Thomas Edison developed the modern lightbulb. He passed an electric current through a thin wire, or **filament**. The electricity heated the filament and caused it to glow brightly. This type of lightbulb is called an **incandescent bulb**.



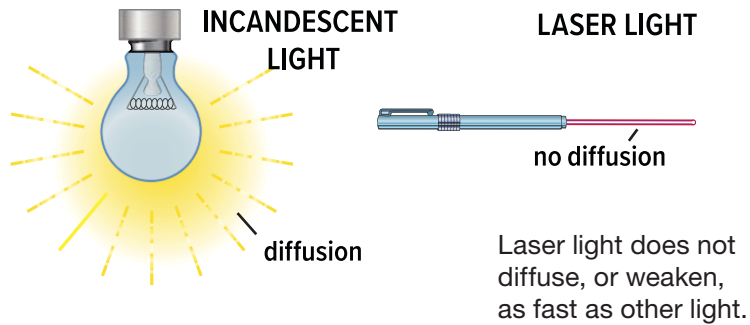
an incandescent bulb



Later, inventors found another way to produce light: by passing electricity through a long glass tube filled with a special gas. The particles of the gas give off photons. The photons strike the inside surface of the tube, which is coated with a special chemical. The chemical glows when struck by the photons and gives off visible light. We call this a **fluorescent bulb**.



Neon lights used for advertising are like fluorescent lights. But neon lights use different gases that produce different-colored light when an electric current passes through them.



Another type of human-made light that people often use is **laser** light. We have learned that light is a useful and powerful source of energy. If this energy can be focused, it can be even more powerful. That is what laser light is: highly focused light. Unlike light from a lightbulb, which spreads out and weakens as it moves away from its source, laser light stays focused and strong.

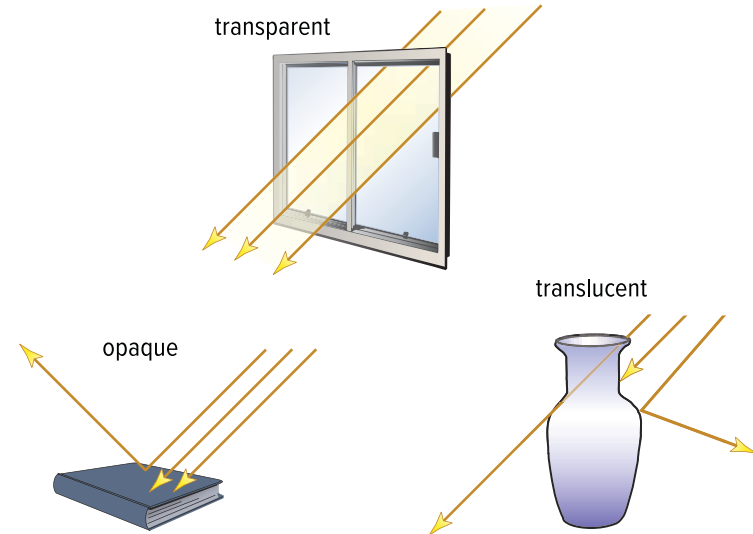
A laser device produces a very narrow and concentrated beam of light. The light itself is all nearly the same wavelength and color. Lasers are used to cut metal, to make incisions during surgery, to read images in scanners and printers, and to read CD and DVD discs.

Word Wise

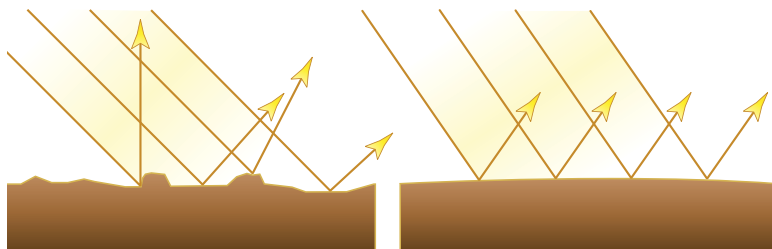
The word *laser* comes from the first letters of the words that describe how it is made: Light Amplification by Stimulated Emission of Radiation.

How Light Interacts with Objects

Light behaves in different ways when it strikes different materials and surfaces. Sometimes light easily passes through. Materials such as clear glass and water only absorb and reflect a small amount of light. These materials are **transparent**. At other times, only some of the light passes through a substance. These materials are **translucent**. Frosted glass and wax paper are examples of translucent materials. Still other materials do not let any light pass through at all. These materials are **opaque**. A brick wall and a book are examples of opaque materials. Opaque materials absorb or reflect the light that strikes them.



Transparent objects let almost all light through, and opaque objects block all light. Translucent objects let some light through, but absorb or reflect the rest.



A rough surface has many angles that scatter light in different directions, while a smooth surface reflects light in a regular pattern.

Reflection

We see most things because of the light that bounces off them. Scientists have observed that light reflecting from a smooth surface bounces off it at the same angle at which it strikes the surface. If the surface is very smooth, like a mirror, light reflects in a regular pattern. Everything looks normal. This explains why standing in front of a smooth, flat mirror allows you to see a perfect image of yourself.

If a surface is uneven, on the other hand, light scatters in many different directions. That's why sunlight sparkles off ocean waves. The waves create many angles on the water's surface, which reflects and scatters light in many directions.



The Moon does not produce its own light. Instead, its surface reflects light from the Sun.

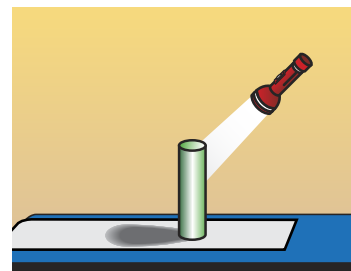


Illustration A

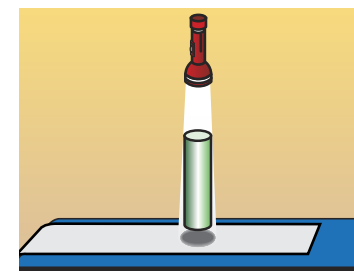


Illustration B

Because opaque objects reflect or absorb light, they cast shadows. A shadow occurs behind an opaque object because light cannot get through the object to the area behind it. The farther an object is from the surface the shadow falls upon, the larger the shadow the object casts (Illustration A). The size of the shadow also depends on the angle and distance of the light. If a light source is far from or directly above an object, the object will cast a small shadow (Illustration B). But if the light is close to or beside the object, its shadow will be bigger. That's why you cast a bigger shadow when the Sun is lower in the sky.



A *solar eclipse* occurs when the Moon passes between Earth and the Sun. A *lunar eclipse* occurs when Earth passes between the Moon and the Sun.



lunar eclipse

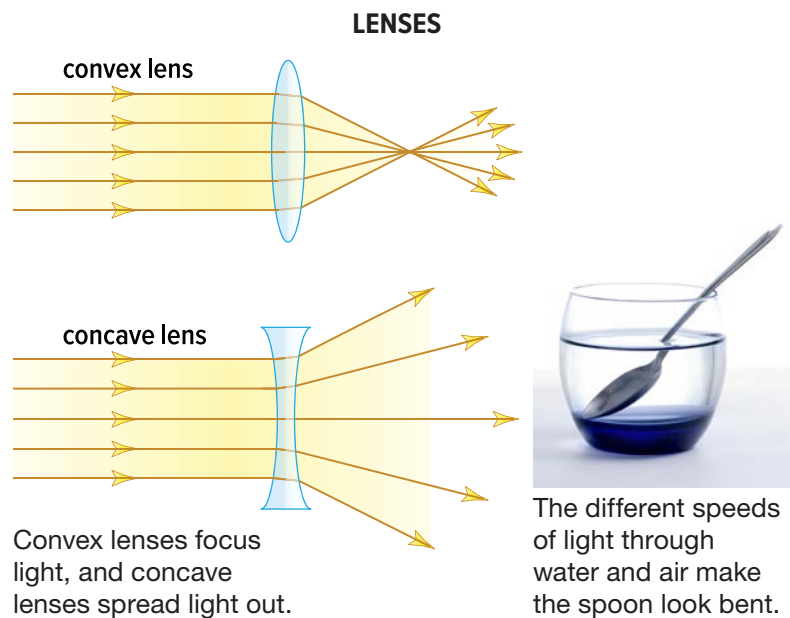


solar eclipse

Refraction

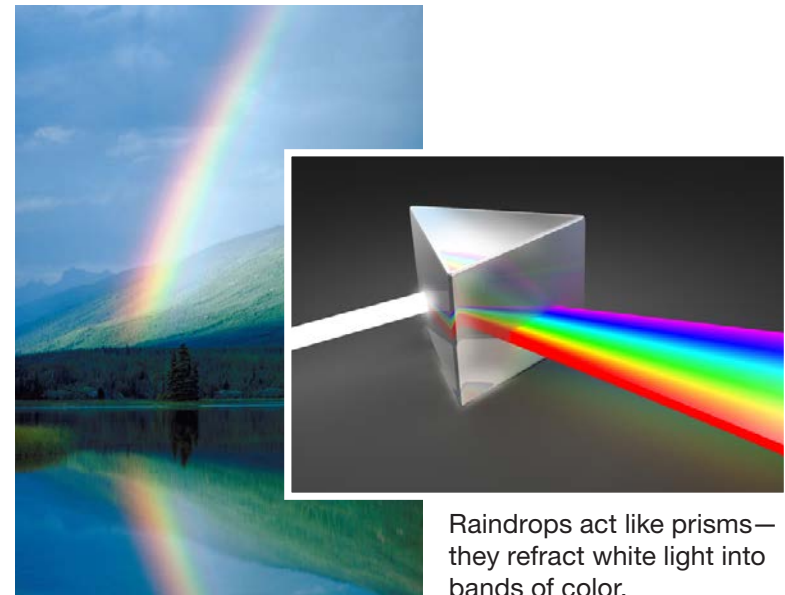
When light passes from one transparent material to another, such as from air to water, it changes speed and bends. This bending of light rays is called **refraction**. Some materials cause more refraction than others. For example, water slows the speed of light more than air does. Look at the photo of the spoon in a glass of water. The changing speed of light as it passes from air to water causes the spoon to appear bent.

Light also refracts when it passes through transparent lenses of different shapes. The diagrams show what happens when light passes through a **convex lens** or a **concave lens**.



All About Color

Visible light is made up of many colors of light. When all these colors are combined, they create what we call **white light**. You can't see the colors of light until they are reflected or refracted to your eyes. Sunlight and visible light from lightbulbs contain all the colors of light. You can separate the colors of light by sending a beam of white light through a prism. The prism refracts the light and separates the colors into a rainbow. Each color of light bends a slightly different amount from the other colors. That's what happens when sunlight passes through raindrops. The raindrops act like prisms, refracting the sunlight into bands of color that you see as a rainbow.



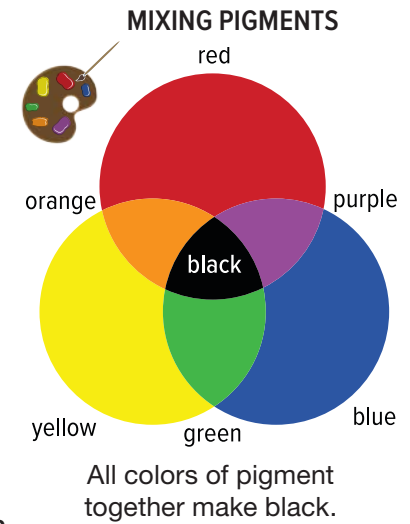


The red tomato absorbs all but red light, and the green tomato absorbs all but green light.

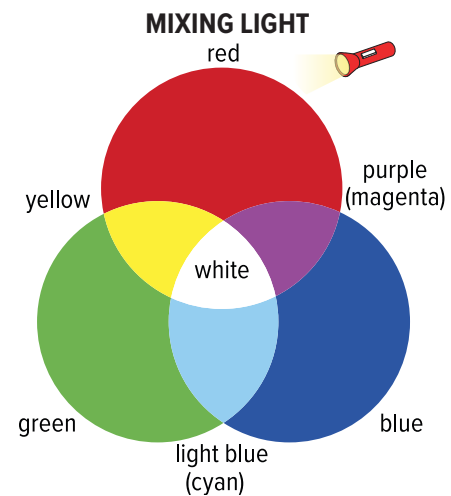
So what causes something to appear a certain color? **Pigments** are what give an object its color. Each pigment absorbs some colors of light and reflects others.

When white light strikes an object, our eyes detect only the colors of light that are reflected. Take a ripe tomato on the vine as an example. It looks red because the surface of a ripe tomato contains red pigment. This pigment absorbs all the colors except red. The reflected red light enters your eyes, and a signal goes to your brain telling you that the tomato is red. If a tomato were not ripe, it would have green pigment rather than red. Therefore, it would absorb the red color along with the other colors and only reflect green. The green would let you know to wait a little longer before picking the tomato.

Paints are pigments that are used to coat objects. Mixtures of pigments can create an enormous variety of colors. The three **primary pigments** are red, blue, and yellow. They can be combined to make all other colors.

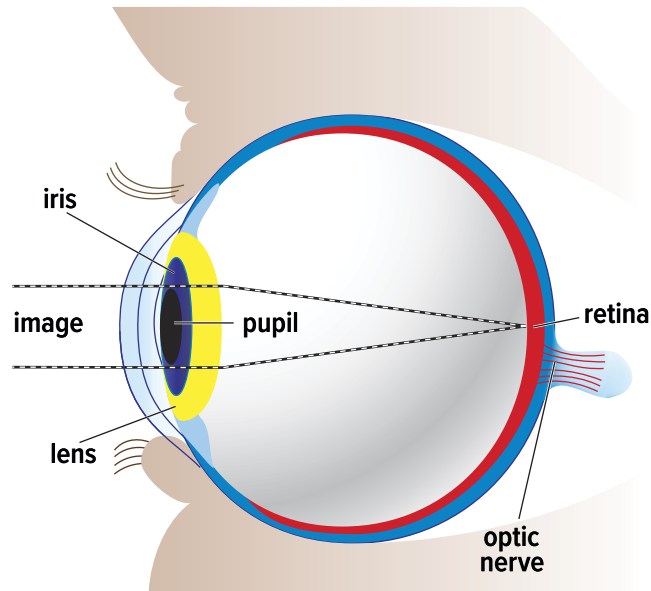


Black is created when all colors of light are absorbed and no color reflects from a surface. An object that appears white is reflecting all colors and absorbing none. Since black surfaces absorb more light energy than white surfaces, black surfaces exposed to light get much warmer than white surfaces.



All colors of light together make white.

Just as pigments can be mixed to create colors, colored light can also be mixed to create other colors of light. However, when you blend all the different colors of light together, you get white, not black.



The lens of the eye focuses light on the retina, which sends a signal to the brain.

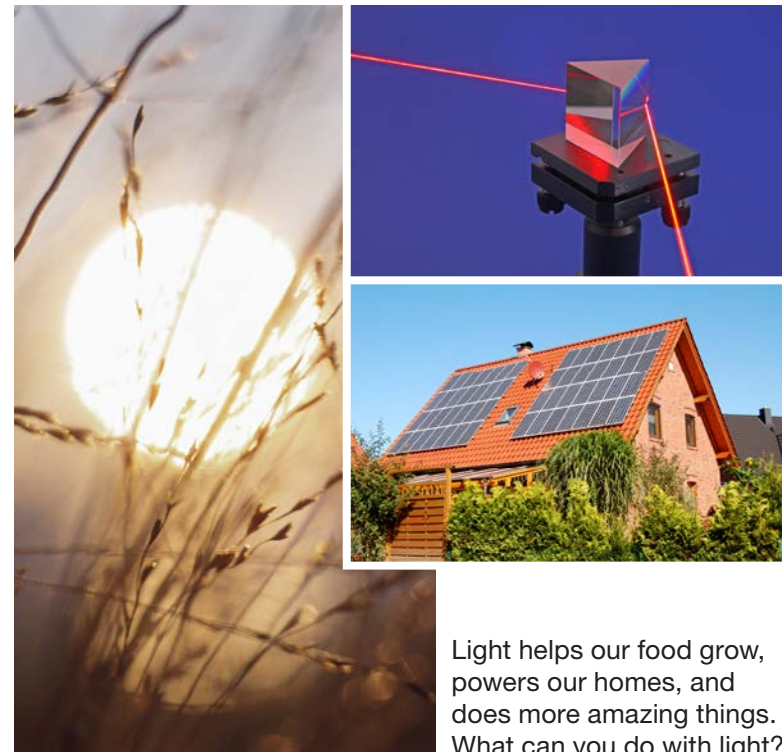
Human Eye

Your eyes are like cameras. They collect light and form images. Each eye, just like a camera, has a small opening through which light enters. This opening is called the pupil. Your eyes also have a colored circle, called the iris, that controls the amount of light entering through the pupil. Light that enters the pupil passes through a clear, convex lens. The lens focuses the light on a filmlike tissue that coats the back of the eye. This tissue, called the **retina**, is sensitive to light. The retina changes the light to a signal and sends it over nerve fibers to the brain. Your brain turns the signal into your view of the world.

Conclusion

You have learned that light is energy and that it can travel on its own, even through the emptiness of space. Light does some amazing things, including bending, refracting, reflecting, and being absorbed. And it does these things incredibly fast. Because it can do all these things, we see the world the way we do.

What are some fascinating things you've learned about light?



Light helps our food grow, powers our homes, and does more amazing things. What can you do with light?

Glossary

concave lens page 17

a lens that is wider at the edges than in the middle and that refracts light rays so they bend outward

convex lens page 17

a lens that is wider in the middle than it is at the edges and that refracts light rays so they come together

fluorescent bulb page 12

a lightbulb in which electricity passes through a gas that reacts with a coating inside the bulb to produce light

frequency page 7

the number of waves that pass a point in a specific length of time

incandescent bulb page 11

a lightbulb in which electricity passes through a wire, which heats up and produces light

laser page 13

intense, focused light of similar wavelengths

light-year page 9

a unit that is the distance that light travels in one year

medium page 8

a substance through which something is carried or transmitted

opaque page 14

blocking all light from passing; not see-through

photon page 6

the smallest particle of light energy

reflection page 15

the return of light or sound as it bounces off something

refraction page 17

the bending of light waves when they pass from one kind of matter to another

retina page 21

a layer of light-sensitive cells at the back of the eye

spectrum page 5

the range of radiant energy, arranged in order of energy or wavelength

translucent page 14

allowing some light through; images are distorted

transparent page 14

allowing all light through; images are not distorted

wavelength page 7

the distance between two consecutive peaks or troughs of a wave

white light page 18

all the colors of light mixed together

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